

group three fossil finds made in the early part of the 19th century. Unlike ordinary reptiles, he said, dinosaurs stood on erect limbs, and their backbones above the hips were fused together. Owen strongly opposed the views of Charles Darwin and used the dinosaurs as an argument against evolution, claiming they were more advanced than a living reptile.

MATTHEW FONTAINE MAURY

1806–1873

A pioneer in oceanography, Matthew Maury was born in Virginia and spent time in the navy before an accident ended his seagoing days. From 1842–1861, he was the first superintendent of the US Naval Observatory. He devoted himself to charting the winds and currents of the North Atlantic by analyzing old ships' logs. In 1855, he published *The Physical Geography of the Sea*, now credited as the first oceanography textbook.

THEODOR SCHWANN

1810–1882

German physiologist Theodor Schwann laid the foundation of cell theory—the identification of the cell as the basic unit of animal structure. He defined the three structural parts of a cell—the wall, nucleus, and fluid content—and said these parts were present in animal as well as plant cells. In 1839, Schwann published the paper *Microscopic Investigation on the Accordance in the Structure and Growth of Animals and Plants*, in which he famously observed that “all living things are composed of cells and cell products.” The Schwann cell—a cell that forms the myelin surrounding the axons of peripheral nerves—was later named after him.

CLAUDE BERNARD

1813–1878

French-born Claude Bernard carried out pioneering work in experimental medicine, including how the body is regulated. His findings provided the foundations for the modern concept of homeostasis—the means by which the body maintains a stable environment that is independent of external changes. He also studied the various processes of digestion—in particular, the roles of the pancreas and liver—and described how chemicals are broken down by digestion into simpler substances and then built up again to make new molecules for body tissues.

WILLIAM THOMSON

1824–1907

Physicist William Thomson, who later became Lord Kelvin, was born in Belfast and raised in Glasgow. His work brought together many areas of physics, and he stated that any physical change is fundamentally a change in energy. He elaborated on the second law of thermodynamics—which states that in any closed system, everything will eventually reach the same temperature—and proposed a temperature scale that was named after him. The Kelvin scale starts at absolute zero (which is $-459^{\circ}\text{F}/-273^{\circ}\text{C}$) but uses the same units as the Celsius scale. He also oversaw the laying of the trans-Atlantic telecommunications cable.

FRIEDRICH AUGUST KEKULE

1829–1896

Born in Darmstadt, Germany, Friedrich August Kekulé was the main originator of the notion of chemical structure. He was educated in Germany, France,

Switzerland, and London before becoming a professor at the University of Bonn. He studied the valency of elements (the maximum number of other atoms they can combine with), and in 1857, he announced that carbon has a valency of 4. His most notable discovery was that the structure of benzene is a symmetrical ring of carbon atoms. Kekulé allegedly “saw” this arrangement after having a dream of a snake biting its own tail.

CARLOS JUAN FINLAY

1833–1915

Cuban physician Carlos Juan Finlay was the first to suggest, in 1881, that yellow fever was transmitted by a species of mosquito. He had studied in Paris from 1860–1861, before returning to Cuba to work as a general practitioner. He became interested in infectious disease and noticed that the *Aedes aegypti* mosquito was often present in houses during yellow fever epidemics. It was 20 years before his ideas were taken seriously. In 1900, researchers on the US Army Yellow Fever Board proved the connection by experimenting on themselves: some bitten by the mosquito went on to develop yellow fever.

AUGUST WEISMANN

1834–1914

German biologist August Weismann is regarded as the most significant evolutionary biologist after Darwin. His lifelong interest in biology began as a boy, and he went on to study medicine. A supporter of Darwin's theory, Weismann proved that inheritance occurs only via sperm and eggs and not from body cells. He did this by cutting off the tails of mice for five generations. All offspring were born with tails, showing that acquired characteristics could not be inherited.